Data Session 14 Key: Inferring the Secret Ballot

84-355 Democracy's Data | Session 14 — Precinct returns, maps, and racial bloc voting | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Framing, Dec 2026. This is the last lab of the term and the payoff of the arc that runs from Session 2 to here. Session 2 asked what the census can measure; Session 3 asked who gets counted as what; Session 7 graded a guess about *who somebody is*; today grades a guess about *what they did*. Same county, same registration file, same move — find the rare case where the truth is recoverable and use it to audit a method that is normally unfalsifiable.

It is also the lab that most directly touches a live case. Section 2 liability turns on the numbers students are about to watch fail.

And the law under it moved on 29 April 2026. *Louisiana v. Callais*, 608 U.S. _____ (2026), left *Gingles* standing but rewrote preconditions two and three: a plaintiff must now "provide an analysis that controls for party affiliation" and show racial bloc voting "that cannot be explained by partisan affiliation," and "simply pointing to inter-party racial polarization proves nothing." Neither method in this lab does that, and the data cannot — the outcome variable is which party's ballot a voter requested. A marked note has been added to both the lab and the brief; the grading bullets below have been updated to match. **The pedagogical payoff is better than before, not worse:** this county is the cleanest possible illustration of why the requirement is hard to meet in the states where Section 2 cases are actually filed.

Every number below was run against the data file.

The lab in one sentence

Ecological regression fits the data almost perfectly, reports that 102% of Black voters took a Democratic ballot, and is wrong by 9.4 points — and the only reason anybody could tell is that Georgia happens to record the answer.

The findings

```r
b <- coef(fit)
bl <- p$pct_black; dm <- p$pct_dem
lo <- pmax(0, (dm - (1 - bl)) / bl); hi <- pmin(1, dm / bl)
c(goodman_black = round(100 * (b[1] + b[2]), 1),
  goodman_white = round(100 * b[1], 1),
  bound_lo      = round(100 * max(lo), 1),
  bound_hi      = round(100 * min(hi), 1),
  truth_black   = round(truth_black, 1),
  truth_white   = round(100 * sum(p$white_dem) / sum(p$white), 1),
  r_squared     = round(summary(fit)$r.squared, 3))
```

```
#> goodman_black.(Intercept) goodman_white.(Intercept)                  bound_lo
#>                   102.100                     3.300                    70.100
#>                  bound_hi               truth_black               truth_white
#>                   100.000                    92.700                     7.100
#>                 r_squared
#>                     0.982
```

- **Truth: 92.7% of Black voters and 7.1% of white voters** requested a Democratic ballot. Near-total polarization.
- **Goodman's regression says 102.1%** — impossible — with an **R^2 of 0.982**.
- **The bounds are [70.1%, 100.0%]**, correct by construction and 29.9 points wide.
- **Homogeneous precinct analysis cannot run at all**: zero precincts are 80% Black. The most Black precinct in the county is 76.6%.
- Every method reached the right conclusion about *polarization*. Every method got the *numbers* wrong. And after *Callais*, none of them reaches the legal conclusion at all — see the framing note above.

The idea to teach

Not "the model was bad." The model was excellent by every diagnostic available to the person running it. That is the point, and it is the hardest one to land.

Three things to get said out loud:

1. **Fitting the aggregate is not recovering the individual.** The regression predicts each precinct's Democratic share to within 4.2 points at worst, 1.7 on average — look at `fit_dem` against `act_dem` in Option C — and still misstates the group rate by 9.4. Predicting the total and decomposing the total are different problems. **This is the ecological fallacy, shown rather than asserted.**
2. **The estimate came from where there is no data.** The Black estimate is the line extended 23.4 points beyond the most Black precinct in the county. Regression will always hand you that number and will never flag it.
3. **The impossibility was a gift.** 102% is self-refuting, so somebody would catch it. Had the truth been 60% and Goodman returned 51%, the report would have been wrong, plausible, and unchallengeable. **Ask the room how they would have caught that.** The honest answer is: they would not have.

The line worth writing on the board: the method was most confident exactly where it had the least evidence.

Option A — Move the homogeneity threshold

Verified output

```r
for (t in c(.80, .70, .60, .50)) {
  h <- p[p$pct_black >= t, ]
  cat(sprintf("  >=%2.0f%%  n=%d  Dem share %s\n", 100 * t, nrow(h),
      if (nrow(h)) sprintf("%.1f%%  (error %+.1f)",
                           100 * sum(h$dem) / sum(h$voters),
                           100 * sum(h$dem) / sum(h$voters) - truth_black)
      else "-- cannot compute"))
}
```

```
#>   >=80%  n=0  Dem share -- cannot compute
#>   >=70%  n=1  Dem share 77.1%  (error -15.6)
#>   >=60%  n=1  Dem share 77.1%  (error -15.6)
#>   >=50%  n=2  Dem share 68.8%  (error -24.0)
```

What a good answer says

The intuitive fix makes it worse, and students should be made to sit with that. Loosening the threshold to get an estimate produces **77.1% at 70% (15.6 points low)** and **68.8% at 50% (24.0 points low)**. Every relaxation admits more white voters, who broke 7.1% Democratic, and drags the estimate down.

3/4: Runs the thresholds, reports that the estimate gets worse as the threshold loosens, and identifies contamination as the cause.

4/4: Notices that 70% and 60% give the *identical* answer, because both capture exactly one precinct — WELLSTON CENTER. The "method" at that point is one polling place, and calling it homogeneous-precinct analysis dresses up a sample size of one.

4/4, best: Turns it into the cross-examination question. There is no principled threshold; 80% is convention. **A witness who reports the 70% threshold and not the 80% one has not lied about anything** — the 80% simply returns nothing. Students who see that the *choice of threshold is where the discretion lives*, and that the discretion is invisible in the final number, have got the lesson.

Option B — Drop the biggest precinct

Verified output

```r
p2 <- p[p$precinct != "WELLSTON CENTER", ]
f2 <- lm(pct_dem ~ pct_black, data = p2, weights = voters); b2 <- coef(f2)
rbind(
  `all 17`      = c(black = 100*(b[1]+b[2]),   white = 100*b[1],
                    R2 = summary(fit)$r.squared, max_black = 100*max(p$pct_black)),
  `drop 1`      = c(black = 100*(b2[1]+b2[2]), white = 100*b2[1],
                    R2 = summary(f2)$r.squared,  max_black = 100*max(p2$pct_black))
) |> round(1)
```

```
#>         black.(Intercept) white.(Intercept) R2 max_black
#> all 17             102.1               3.3  1      76.6
#> drop 1             106.8               1.7  1      50.0
```

What a good answer says

Removing one precinct out of 17 moves the Black estimate **from 102.1% to 106.8%** — further into impossibility — and the R^2 stays at **0.966**.

3/4: Reports the shift and notes the fit is still excellent.

4/4: Connects it to the extrapolation. Without WELLSTON the most Black precinct is **50.0%**, so the estimate is now the line extended *50* points past the data. The estimate got worse in exact proportion to how far it was extrapolated — which tells you the number was never really an estimate of Black voters at all, but a property of the line's slope.

4/4, best: Asks the fragility question the prompt sets up. **A fifty-year-old courtroom standard, in this county, rests on one polling place.** A student who says they would want to know, for any RPV report, how many precincts anchor the minority end — and that this is never reported — is thinking like a reviewer.

Option C — Predict, then check

Verified output

```r
p$true_blk <- 100 * p$black_dem / p$black
p$fit_dem  <- 100 * fitted(fit)
p$act_dem  <- 100 * p$pct_dem
o <- p[order(p$pct_black), c("precinct","pct_black","act_dem","fit_dem","true_blk")]
print(o, row.names = FALSE, digits = 3)
```

```
#>                     precinct pct_black act_dem fit_dem true_blk
#>                         HEFS     0.109    11.5    14.1     85.4
#>                         HAFS     0.114    13.1    14.6     86.8
#>                         ROZR     0.140    16.3    17.1     92.4
#>                          BMS     0.192    23.3    22.3     91.8
#>                         CGTC     0.225    25.7    25.5     92.4
#>                         ANNX     0.236    27.6    26.7     91.3
#>                          PEC     0.240    27.3    27.0     90.5
#>                          VHS     0.252    27.0    28.2     93.6
#>                          VFW     0.256    32.7    28.6     93.2
#>                         MCMS     0.299    30.6    32.8     89.1
#>                         TWPK     0.301    34.7    33.1     93.3
#>  NORTH HOUSTON SPORT COMPLEX     0.314    32.5    34.3     91.0
#>                         CENT     0.344    37.2    37.3     93.1
#>                         HHPC     0.359    42.1    38.8     92.4
#>                         FMMS     0.369    38.2    39.7     93.0
#>                          TMS     0.500    56.1    52.7     95.7
#>              WELLSTON CENTER     0.766    77.1    79.0     96.3
```

```r
cat(sprintf("\nTRUE Black Democratic rate ranges %.1f%% to %.1f%% - spread of %.1f points.\n",
            min(p$true_blk), max(p$true_blk), max(p$true_blk) - min(p$true_blk)))
```

```
#>
#> TRUE Black Democratic rate ranges 85.4% to 96.3% - spread of 10.9 points.
```

What a good answer says

This is the best of the five options and worth steering strong students toward. It exposes the actual mechanism.

3/4: Notices `fit_dem` tracks `act_dem` closely — the model predicts precinct outcomes well — while the group estimate is wrong.

4/4: Reads the `true_blk` column and sees that **Black Democratic support is not constant across precincts: it runs from 85.4% in HEFS to 96.3% in WELLSTON CENTER, a spread of 10.9 points.** Goodman's regression assumes that rate is the *same everywhere* — that is precisely what a single intercept and slope encode. The assumption is false in the data, and we can prove it only because we have the column.

4/4, best: Notices the pattern in the violation. Black voters are *more* Democratic in precincts with more Black voters. That correlation between group behaviour and group share is exactly the condition under which ecological regression is known to break, and it is not detectable from aggregates — it looks like a slightly steeper line. A student who gets here has independently rediscovered why King wrote the book.

Option D — Would it transfer?

No output; graded on reasoning.

3/4: Lists real differences between a May primary and a November general — smaller electorate, more partisan, different composition.

4/4: Names the specific bridging assumption. To carry today's validation to November you must assume the *method's error* behaves the same in both, which is a claim about the geography–behaviour correlation, not about turnout. Nothing we did tests that.

4/4, best: Observes that the validation is available only in states that record race *and* run partisan primaries, which is a small and unrepresentative set of mostly Southern states — the same places most likely to be in Section 2 litigation. **The evidence about whether these methods work is drawn from exactly the counties where they are most often used, and that is either reassuring or circular depending on how you look at it.** Either answer, argued, is a 4.

Option E — Own question

Grade on whether the question could be answered with the columns present. Common good ones: does precinct size predict error; does the Republican side mirror the Democratic side (it must, with two parties — worth letting a student discover); what happens with an unweighted regression (white 3.0%, black 103.0% — the weighting is not the problem).

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is refusing to conclude "the method is broken." Every method got the polarization answer right. The correct reading is narrower and harder: the methods are reliable for *direction* and unreliable for *magnitude*, and until this year Section 2 mostly needed direction — which was the strongest available defence of fifty years of practice, and students should be able to state it before they attack it.
- **Do not let a student stop there now.** *Callais* took away the second half of that defence: direction alone is no longer sufficient, because the direction has to be net of party. A student who notices that the lab's outcome variable *is* party, and that this makes the county an illustration of the problem rather than a solution to it, has done better than the document did before the note was added. **A 4 either way** — the pre-*Callais* defence stated accurately as history is still a 4.
- **A 2** is "statistics can prove anything" or "the ecological fallacy means you can't do this." Both skip the work.
- **Reward anyone who is bothered that we validated on the easy case.** Polarization here is near-total; the errors did not matter. That is the right thing to be uneasy about.

Anticipated questions

- *"Why not just use King's method?"* — We could, and it does better here. It is still a model, it still cannot be checked in the counties where it is actually used, and its assumptions are harder to state than Goodman's rather than weaker. Worth being straight about: this lab is not an argument for any particular method, it is an argument about what validation is available.
- *"Isn't 102% just a rounding problem?"* — No. It is 9.4 points past the truth and 2.1 points past possibility. Say plainly that a percentage over 100 means an assumption failed, not that a number was rounded.
- *"If race is on the file, why infer at all?"* — Because race is on the file in a handful of states, and how people **voted** is nowhere. Today's shortcut is the party-ballot request, which exists only in partisan primaries. In November nobody knows anything.
- *"Did we just prove Houston County is polarized?"* — We showed its **May primary electorate** was. Do not let this become a finding about the county; it is a finding about a method.
- *"Does* Callais *make this lab pointless?"* — The opposite, and say so. The statistical content is untouched: ecological inference is still the only route from precinct aggregates to group behaviour, and Steps 6–9 still show it failing. What changed is that the estimate no longer carries the case by itself. If a student asks what a party-controlled analysis would look like here, the honest answer is that the Court required one and did not specify it, and that in a partisan-primary file the natural fixes all reduce to comparing Black and white voters *inside* one party's primary — which is what *Gingles* itself rested on, 478 U.S., at 59. We do not run that today.
- *"Is 17 precincts too few?"* — It is on the small side but entirely ordinary for a county-level Section 2 case. Do not let students dismiss the result as a small-sample artefact: the failure is structural — extrapolation past the data and a false constancy assumption — and more precincts of the same shape would not fix either.

Connections

- **Session 7 (surnames and BISG)** — the direct predecessor and the same county. Both labs grade a normally-ungradeable inference against Georgia's registration data. Say so explicitly; the pairing is the argument.
- **Session 3 (the ACS and §203)** — the statute this evidence is produced for.

- **Session 2 (the decennial census)** — where the denominators in every RPV case come from.
- **Session 8 (CPS turnout)** — the term's first "the measurement is not the thing" lab, and the first appearance of a number that looked fine and was not.
- **None of this material is on a test** — all three tests cover the textbook only, and Test 3 (same day as the wrap) is Sides et al. Ch. 9–11. Say so early. The risk is that students read "not on the test" as "not important" in the four weeks the labs were actually built for, so the board post is the whole assessment here and worth treating as such.